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Nested relation extraction via self-contrastive learning guided by structure and semantic similarity

Chengcheng Mai, Kaiwen Luo, Yuxiang Wang, Ziyan Peng, Yu Chen, Chunfeng Yuan, Yihua Huang *

State Key Laboratory for Novel Software Technology, Nanjing University, Nanjing, 210023, China
Department of Computer Science and Technology, Nanjing University, Nanjing, 210023, China

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ABSTRACT

The conventional Relation Extraction (RE) task involves identifying whether relations exist between two entities in a given sentence and determining their relation types. However, the complexity of practical application scenarios and the flexibility of natural language demand the ability to extract nested relations, i.e., the recognized relation triples may be components of the higher-level relations. Previous studies have highlighted several challenges that affect the nested RE task, including the lack of abundant labeled data, inappropriate neural networks, and underutilization of the nested relation structures. To address these issues, we formalize the nested RE task and propose a hierarchical neural network to iteratively identify the nested relations between entities and relation triples in a layer by layer manner. Moreover, a novel self-contrastive learning optimization strategy is presented to adapt our method to low-data settings by fully exploiting the constraints due to the nested structure and semantic similarity between paired input sentences. Our method outperformed the state-of-the-art baseline methods in extensive experiments, and ablation experiments verified the effectiveness of the proposed self-contrastive learning optimization strategy.

1. Introduction

The conventional relation extraction (RE) task involves determining whether relations exist between given entities and identifying their relation types, and it plays a central role in various downstream natural language processing (NLP) tasks, such as knowledge graph construction (Huang, Mao, Yang, Zhu, & Long, 2020; Xu & Li, 2019; Zhang, Deng, et al., 2020), text summarization (Safavi et al., 2019; Wei, Wang, & Wang, 2020), and question answering (Li et al., 2019; Qiu, Wang, Jin, & Zhang, 2020; Qiu et al., 2018).

However, due to the complexity of natural language, many complicated relation structures may be identified in practical application scenarios, including formula extraction from texts (Cao, Hong, Li, & Luo, 2021), fact checking (Cao, Chen, Li, & Luo, 2019), and causality analysis (Chen, Cao, & Luo, 2020). Depending on whether the relations are nested or not, RE tasks can be further subdivided into two subtasks comprising flat RE (Geng, Chen, Han, Lu, & Li, 2020; Kuang et al., 2020; Qiao et al., 2022) and nested RE (Cao et al., 2019).

The flat RE task involves categorizing the types of relations between head and tail entities. However, the nested RE task considered in the present study has a more complicated structure, where it is necessary to identify and extract the relations between entities but also the relations between entities and relation triples, and the relations between relation triples. The existing RE methods are not suitable for dealing with this problem. To some extent, the nested RE task has a more general form than the flat RE task, and the nested RE task is compatible with the flat RE task. Therefore, the nested RE task is the focus of the present study. Fig. 1 shows examples of the two types of RE subtasks.

In contrast to flat RE, not only are the entities participants in relations in the nested RE task, but the lower-level relation triples can also be components of the higher-level relations. As illustrated by the example in Fig. 1(b), a nested relation extractor that can automatically parse the entities and logical dependency relations from input example sentences (which are excerpted from the acceptance requirements of preferential policies issued by the government annually) is desired by many business enterprises to guide them to declare policies and cash the corresponding financial subsidies. By contrast, flat RE methods can only extract relation triples with a flat structure, such as R_{attr} <"*revenue*", @, "2019"> in the first layer in Fig. 1(b) ***Sen.3***, and they cannot deal with nested relations, such as R_{eql} < R_{attr} <"*revenue*", @, "2019">, =, "$ 2 *million*">.

* Corresponding author at: State Key Laboratory for Novel Software Technology, Nanjing University, Nanjing, 210023, China.
E-mail address: yhuang@nju.edu.cn (Y. Huang).

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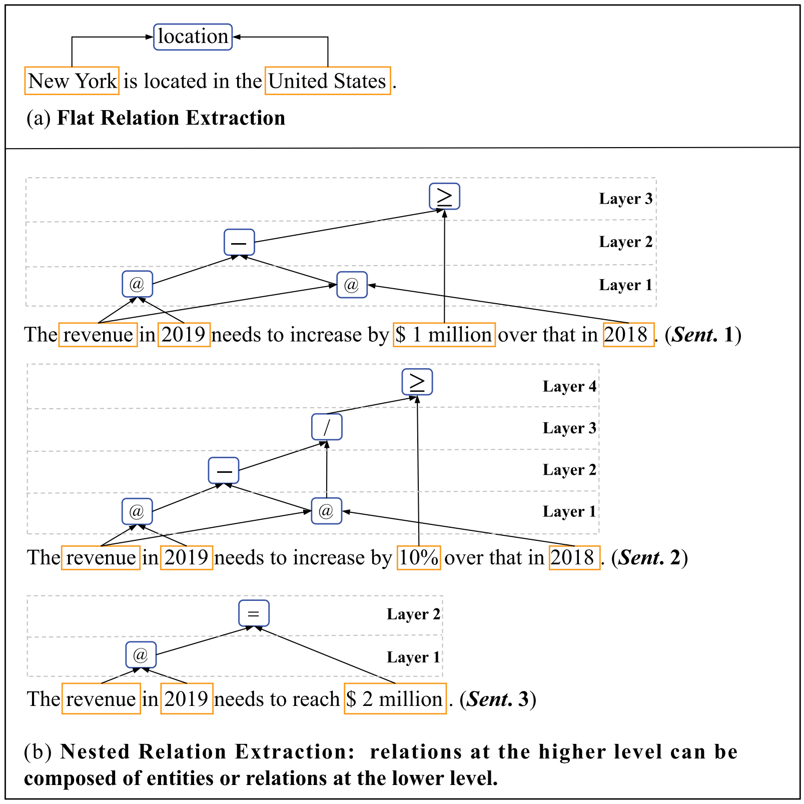


Fig. 1. Example of subdivided categories in relation extraction subtasks. Symbols "@" , "=", "−", "/" an ">" represent the "*Attribute*", "*Equal to*", "*Subtraction*", "*Division*" and "*Greater than*" relations defined in Section 5.1, respectively. ***Sent***. # is short for the sentence number.

Several studies have suggested the great value of nested RE in scientific research and practical applications but research into nested RE is still in its infancy and many challenges remain. In particular, the existing RE models were mostly designed for identifying relations between two entities, and they are not suitable for nested structures. Cao et al. (2019) expressed the nested structure as a directed acyclic graph (DAG) and proposed an iterative neural network (INN) based on the long short-term memory (LSTM) (Hochreiter & Schmidhuber, 1997) model to extract the nested structures of relations in a layer by layer manner. Other challenges that need to be solved include the following: **(1)** when the nested RE model is applied to a specific application field, abundant labeled data are probably not available considering the time-consuming and labor-intensive data annotation process, and thus the model must be adapted to low-data settings; and **(2)** most previous studies focused on the semantic dependencies between entities whereas they ignored the unique structural characteristics of the nested relations, which actually plays an important role in learning the constraints between semantics and structures for improving the nested RE recognition performance.

To address the low-data dilemma, we utilize the contrastive learning method as the fundamental model, which has been proven effective in few-shot learning, in order to learn feature representations that can significantly distinguish positive and negative instances by gathering similar neighbors together and pushing non-neighbors apart (Giorgi, Nitski, Wang, & Bader, 2021; Hadsell, Chopra, & LeCun, 2006; Kalantidis, Sariyildiz, Pion, Weinzaepfel, & Larlus, 2020; Tian et al., 2020; Ye et al., 2021). However, in the nested RE task, the construction of positive and negative instances is more complicated because multiple relations are present in one sentence, and both entities and relations can be the constituent elements of nested relation triples. Some relation types are the same in the compared sentence pairs but the elements constituting these relations may be different, which makes it difficult to generate a completely positive or negative contrastive instance according to the conventional contrast learning paradigm. For example, as illustrated in Fig. 1(b) the compared sentence pair (***Sent.*** 1, ***Sent.*** 2) has three identical entities and four identical flat or nested relations, but ***Sent.*** 2 has one unique nested relation denoted by "/", and thus the two compared sentences contain both identical and different relation triples and nested structures.

To solve the uncertainty problem regarding positive and negative instances, we reformulate the existing contrastive learning method and propose a novel **S**elf-**C**ontrastive **L**earning (**SCL**) strategy inspired by an interesting phenomenon observed in Fig. 1(b), i.e., "***when the structures of a sentence pair are more similar, the semantics are more similar, and vice versa***", which can directly learn the discriminative representations by minimizing the distance between both the structural and semantic similarities of two compared input sentences, rather than artificially fabricating positive and negative instances.

Thus, we conduct the SCL strategy based on an INN comprising the Transformer model (Vaswani et al., 2017), which takes the hidden states and types of the constituent elements (entities and relations from the lower layer, as described by Cao et al. (2019)) as multi-feature input sequences to identify more new relation triples from the upper layer until no new relations can be found. To the best of our knowledge, this is the first method to extract nested relations under the SCL framework by considering both the structural and semantic similarities. The validity of this concept was demonstrated in ablation experiments.

The main contributions of our work can be summarized as follows:

- We formally define the nested RE task and present an INN based on Transformer with multi-feature sequences as inputs to extract the nested relations in a layer by layer manner.
- A novel SCL method is proposed to strengthen our method for performing the nested RE task in low-data settings by exploiting the structural and semantic similarities, and constraints between paired input samples. To the best of our knowledge, we are the first to combine structure and semantic information for the nested RE task with self-contrastive loss, thereby extending the existing contrastive learning methods, and this method is suitable for situations where partial similarities exist between compared sentence pairs.
- Extensive experiments were conducted to demonstrate that our proposed SCL-nestedRE method can outperform the existing state-of-the-art (SOTA) baseline methods in both resource-rich and low-data settings. Ablation experiments also verified the effectiveness of our proposed iterative Transformer architecture and SCL-based optimization strategy.

The remainder of this paper is organized as follows. In Section 2, we discuss related work. In Section 3, we provide definitions for the nested RE task. In Section 4, we describe our method, and the experimental results are analyzed in Sections 5 and 6. We give our conclusions in the final section.

2. Related work

Relation Extraction (RE). Recognizing relation triples in unstructured text is an essential part of the automatic information extraction process, and it can fundamentally free people from conducting complicated data analysis tasks, such as knowledge graph construction (Shi, Xiao, Quan, Lei, & Niu, 2021) and knowledge reasoning (Chen, Jia, Ding, Shen, & Xiang, 2020; Huang, Lei, & Feng, 2021). The different application scenarios demand more in-depth and detailed studies of the RE task, which requires categorization into fine-grained subtasks comprising flat RE and nested RE, according to whether the constituent elements of relation triples are nested or not.

Flat RE refers to the conventional RE task where the constituent elements are only two non-overlapping entities. Previous studies (Gormley, Yu, & Dredze, 2015; Zeng, Liu, Lai, Zhou, & Zhao, 2014) performed this task using a pipeline approach by first extracting all possible entities with named entity recognition methods and then applying a relation classification algorithm based on pairs of these entities. Alt, Hübner, and Hennig (2019) proposed a Transformer-based RE model. The proposed model uses deep semantic representations that are fine tuned with Transformer as inputs for the relation classifier, instead of using explicit linguistic features, such as part-of-speech tags, syntactic features, and morphological features. The Transformer-based model is advantageous for capturing the long-range dependencies between input words and obtaining contextualized semantic representations. Ding et al. (2021) presented a prototype-based approach for RE. Each prototype serves as the semantic center to cluster different sentences with the same relation in order to learn the essential semantics of relations. The advantages of the prototype-based approach are that it can ensure both intra-class compactness and inter-cluster separability to obtain interpretable and robust representations of relations. Guo, Wang, and Gao (2022) proposed using a dependency position encoding to incorporate syntactic dependency information into the RE method for capturing the relations between entities. In contrast to previously proposed dependency-based methods, this method considers the importance of different dependency types and assigns different weights to different dependency types by using a self-attention mechanism to filter the confusion in syntactic dependency information. Furthermore, to avoid the noises caused by the introduction of extra dependency trees, Tian, Song, and Xia (2022) proposed two kinds of pre-training tasks, namely the masked dependency connections recovery (prediction) and the masked dependency types recovery (prediction) task, to introduce the dependency information between words and entities into the encoder during the pre-training stage. Attention-guided graph convolutional networks (Guo, Zhang, & Lu, 2019) have also been applied to encode and extract relations between entity nodes by transforming the syntactic dependency tree into a fully connected graph. To fully exploit the domain knowledge of entity for relation extraction, Hong et al. (2021) used knowledge representation learning as an auxiliary task to explicitly encode the context and direction features of entity pairs under the multi-task learning paradigm. Specifically, in addition to predicting relations according to the head and tail entities and their contexts, the author also used TransE (Bordes, Usunier, Garcia-Duran, Weston, & Yakhnenko, 2013), TransH (Wang, Zhang, Feng, & Chen, 2014) and DistMult (Yang, Yih, He, Gao, & Deng, 2015) models respectively to map the representations of entities and relations into the unified vector space by making the semantic distance between positive samples closer and the semantic distance between negative samples further.

To avoid the error propagation problem of pipeline-based methods, another approach involves jointly extracting entities and flat relations, as applied in several studies (Wang & Lu, 2020; Zheng et al., 2017). Chen et al. (2021) incorporated more high-order semantic information into the relation task, such as features of part-of-speech tags, n-grams, and relative position information for input words, to increase the discriminability of the semantic representations for neural networks. Zhang, Zhang, and Fu (2017) used a method based on table filling to extract relations and introduced additional syntactic constraints into their model to reduce the interference effects caused by noisy data on the RE model, thereby further improving the robustness of the flat RE model. To further improve the performance of the RE task, distant supervision RE methods have also been widely used. Zhou, Pan, et al. (2021) proposed a self-selective attention method to encode the entities and input words for learning better semantic embeddings. To handle the noise from distant corpora or knowledge bases, Zhang, Liu, et al. (2020) presented a question–answering based method to reduce the noise at the word level and designed a multi-instance learning strategy for sentence-level noises. Yan, Zhang, Fu, Zhang, and Wei (2021) proposed a partition filter network model to solve the problem that in the joint entity and relation extraction, the features extracted for two tasks are not directly related to each other. Firstly, the neurons in the encoder are subdivided into the entity, relation, and shared encoding partitions using the designed the entity and relation gates. Then, three kinds of memory blocks are presented to filter information irrelevant to the entity or relation extraction task, for improving the performance of joint entity and relationship extraction. These sophisticated joint models have obtained promising results in flat RE, but more complex relation structures need to be identified, such as nested RE.

Nested RE further extends the scope of the constituent elements of relations, including the relation triples from lower layers, which is more appropriate for real-world applications, but this approach also brings more challenges. To parse the nested structure, Cao et al. (2019) proposed an INN based on LSTM to extract relations based on the lower layer, and the recognized relation triples are then used as the new candidate constituent elements to generate new relations based on the higher layer. Nested structures are quite common in real-world scenarios, and thus they have inspired many related applications. In the math word problem (MWP) field, the key to solve math problems described in a natural language is the method used to recognize nested relations between quantities and operators (Cao et al., 2021). In the field of causality mining, Chen, Cao, and Luo (2020) found that complex nested causalities can be converted into multiple pairwise causality structures by iteratively connecting the effects of child relations with the causes from their parent relations. Thus, a nested causality detection model was presented to discover correlations between events from financial statements. Zhou, Ma, et al. (2021) also proposed a nested causality extraction method for traffic accident cause analysis based on a question–answering template. In this method, the position of the trigger word representing the "cause" is first identified and the "cause" trigger word and question–answering template are then combined into a question based on a self-defined question–answering template. The question is concatenated with the original input sentence and the obtained sentence is input into the model to identify the location of the trigger word representing the "effect" caused by the trigger word of the "cause", and the causal relation triples are extracted. By iteratively executing this process, the nested causality relations can be extracted. However, in order to simplify the representation of causal relation triples, the two methods described above only use the "effect" trigger word to represent the whole causality triple, which would lead to ambiguity when dealing with multiple nested causality relations with the same "effect" trigger word.

Despite the huge number of potential applications, nested RE has rarely been explored because of the lack of labeled corpora, especially when applied to a specific application field. Therefore, the existing nested RE methods need to be optimized to adapt to low-data settings.

Contrastive learning. Contrastive learning is a popular recent paradigm based on data augmentation for alleviating the low-data problem, where the aim is to acquire the most representative feature representations by distinguishing positive and negative samples in order to improve the performance of downstream tasks (Giorgi et al., 2021; Lin et al., 2021; Yang et al., 2022, 2021; Zhang et al., 2021). Contrastive learning first attracted much attention in the field of computational vision and it has achieved great success in representation learning. Lin et al. (2021) proposed a contrastive-learning-based method for learning informative representations through different views of data, which can also recover missing views from data by minimizing the conditional entropy loss. To align the intrinsic multiple views of data to alleviate noise in multi-view clustering, Yang et al. (2021) presented a noise-robust contrastive learning method to prevent false negatives from dominating the optimization process, which uses aligned samples as positives and constructs negatives by randomly pairing cross-view samples. Contrastive learning was subsequently applied widely in the field of NLP. Word deletion, span deletion, reordering, synonym substitution, and back-translation are generally used as data augmentation strategies for generating positive or negative instances in NLP tasks (Ding et al., 2020; Wu et al., 2020). However, data augmentation for NLP is still difficult due to its discrete nature. Thus, Gao, Yao, and Chen (2021) showed that dropout can also be applied as a data augmentation method for hidden states of continuous word or sentence embeddings, and this method empirically outperforms the discrete augmentation method. In addition, most previous studies focused on the semantic similarities between positive and negative samples, whereas they ignored structural similarities in the nested RE task, which are actually crucial for recognizing the dependency relations between entities and nested relations.

To the best of our knowledge, we are the first to combine the structure and semantic similarity simultaneously in the contrastive learning paradigm. Furthermore, in the nested RE task, a sentence contains multiple relation triples, so there is a high probability that the two compared sentences are partially similar and thus it is difficult to define the boundary between the positive and negative instances. Therefore, we propose a novel SCL method, instead of artificially constructing positive and negative instances.

3. Preliminaries

We first define the nested RE task and then introduce the constraints between the constituent elements of nested relations for candidate relation filtration.

3.1. Definition of the nested RE task

The nested RE task involves identifying and categorizing the predefined relations between constituent elements. In contrast to the conventional RE task, the constituent elements in the nested RE task are not limited to entities but instead they also can be relation triples. The related concepts are defined as follows.

Definition 1 (***Entity***)**.** An entity is composed of a sub-sequence of the entire input word sequence $\{w_1, \ldots, w_i, \ldots, w_n\}$, denoted as entity $:= \mathrm{ent}(w_i, \ldots, w_j)$ $(1 \leq i \leq j \leq n)$, with the entity type predefined in the set of entity types denoted as $E = \{ent_{t_1}, \ldots, ent_{t_i}, \ldots, ent_{t_m}\}$, where n represents the length of the input sequence.

In Fig. 1(b), entities exist such as *Econ*("*revenue*), *EconVal* ("\$ 1 *million*"), and *Time*("2019"), where "*Econ*", "*EconVal*" and "*Time*" represent the entity types.

Definition 2 (***Nested Relation***)**.** The nested relation r^l in the lth layer is defined in the form of a triple: $r^l := < elem_l, RelType, elem_r >$, where $elem_l$ and $elem_r$ represent the left and right constituent elements of relation r^l, respectively. *RelType* represents the relation type predefined in the set of the entire relation types denoted as $R = \{r_1, \ldots, r_i, \ldots, r_t\}$. L is the number of the layers of the nested relations and $l \in \{1, \ldots, L\}$ denotes that r^l is in the lth layer of the nested structure.

Definition 3 (***Element***)**.** The constituent elements in nested relation r^l can be entities or nested relations, which are defined in nested form as: $elem_{\{left,right\}} \in \{entity, r^{l-1}\}, (1 \leq l \leq L)$, where L represents the number of the layers of the nested relations.

For example, as illustrated in ***Sent.*** **3** in Fig. 1(b), entity "*revenue*", "2019" and relation type "@" which denotes the attribution relation, form a flat relation triplet in the form of R_{attr}<"*revenue*", @, "2019">, which means "the revenue in 2019". On this basis, relation R_{attr} and entity "\$ 2 *million*" further form a nested relation triple, with relation *Eql*("="), in the form of R_{eql}< R_{attr}<"*revenue*", @, "2019">, =, "\$ 2 *million*">, which means "the revenue in 2019 is \$ 2 million". In nested relation R_{eql}, the constituent elements are flat relation R_{attr} and entity "\$ 2 *million*". In the process above, the current extracted relation R_{attr} can also be used as an element to identify subsequent relations in the higher layer. Therefore, these relations have a nested structure with multiple layers.

3.2. Candidate relation filtration based on constraints between elements

Formulating the constraints between the constituent elements is a necessary step before performing the nested RE task, which

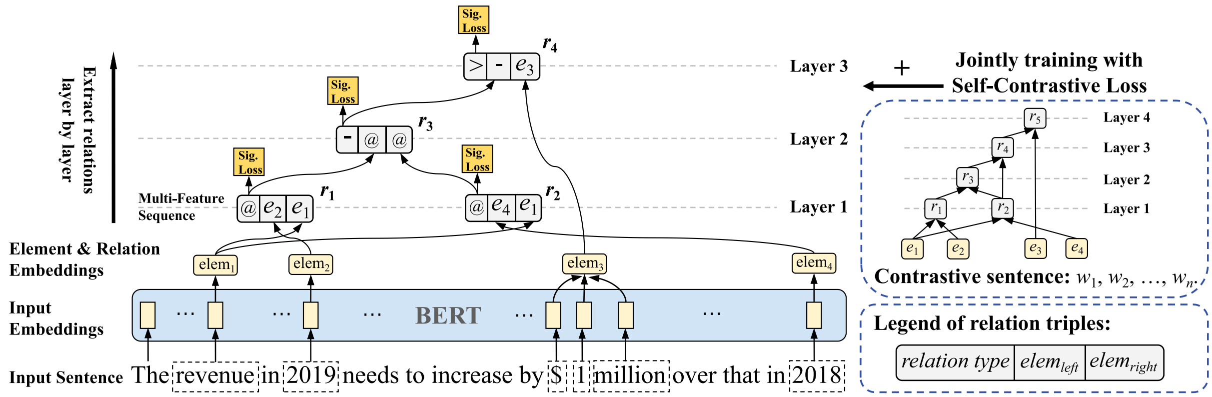


Fig. 2. Overview of our SCL-nestedRE method.

can help to reduce the search space when pairing elements into triples by constraining the element pairs that can or cannot form a relation.

First, we define the constraints between relation elements and then present a candidate relation filtration algorithm for element pairs.

Algorithm 1 Retrieval method for candidate relation set filtration

Input: element pair $elemPair_j$ =< $elem_j^{left}$,$elem_j^{right}$>,$1 \leq j \leq N$, Constraints i for relation r_i, $1 \leq i \leq t$, N is the number of all input element pairs, t is the serial number of relation types.
Output: Candidate relation set S^r for all input element pairs

```
set 𝒮^r = ∅;
for j = 1 to N do
    set 𝒮_j^r = ∅; //relation set for elemPair_j
    for i = 1 to t do
        // t denotes the number of all relations
        if elem_j^left ∈ 𝒮_i^{elem_l} and elem_j^right ∈ 𝒮_i^{elem_r} then
            // 𝒮_i^{elem_l}, 𝒮_i^{elem_r} refers to left and right element set of relation r_i
            add r_i into 𝒮_j^r: 𝒮_j^r = 𝒮_j^r ⋃ {r_i}
        end if
    end for
    add 𝒮_j^r into 𝒮^r: 𝒮^r = 𝒮^r ⋃ 𝒮_j^r
end for
Return 𝒮^r = 𝒮^r_{j=[1,...,N]}
```

3.2.1. Definition of constraints between elements

Definition 4 (***Constraints***)**.** Given a relation r_i, the constraint on the value range of the left and right constituent elements of r_i can be formalized as a mapping function $f_{const.}$:

$$Const.i: \ r_i \xrightarrow{f_{const.}} \mathcal{S}_{r_i}^{elem_l} \times \mathcal{S}_{r_i}^{elem_r}$$

where $\mathcal{S}^{elem_l}, \mathcal{S}^{elem_r} \in \{E \bigcup R\}$ represents the possible type set of the elements for relation r_i, and E and R represent the sets of entity and relation types, respectively.

Only the elements that satisfy the constraints can be used as the left and right elements of the relations. The constraint mapping function is manually formulated according to the requirements of specific application scenarios.

3.2.2. Candidate relation filtration

After collecting the constraints for all relations, $R = r_1, \ldots, r_i, \ldots, r_t$, based on their constituent elements, all possible candidate relations for a given pair of elements, $elemPair_j$=< $elem_j^{left}$, $elem_j^{right}$>, can be deduced by reverse-retrieving and matching the element set in these constraints. If $elem_{left} \in \mathcal{S}^{elem_l}$ and $elem_{right} \in \mathcal{S}^{elem_r}$, then the relation $r_i, (1 \leq i \leq t)$, will be annotated as the candidate relations for the element pair. The detailed candidate relation filtration algorithm is described in Algorithm 1 as follows.

For example, as illustrated in ***Sent.* 1** in Fig. 1(b), the process for generating the possible relation set can be subdivided into three steps.

Step 1. Construct the set of entities and relations. $E = \{Econ, EconVal, Time\}$, $R = \{@, \text{-}, \geq\}$.

Step 2. Formulate the type constraints for elements in relations:

$$Const.1: \ r_1 = \text{“@"} \xrightarrow{f_{const.}} \{Econ, Time\} \times \{Econ\}$$

$$Const.2: \ r_2 = \text{“} - \text{"} \xrightarrow{f_{const.}} \{Econ, @\} \times \{Econ, @\}$$

$$Const.3: \ r_3 = \text{“} \geq \text{"} \xrightarrow{f_{const.}} \{Econ, Time, -\} \times \{Econ, EconValue\}$$

Step 3. Generate the candidate relation set. We obtain the candidate relation set $\{@, \geq\}$ by retrieving and comparing the constraints formulated in **Step 2.**:

For a given element pair $< Time(\text{“2019”}), Econ(\text{“}revenue\text{”})>$ in ***Sent.* 1**, the first and second elements are contained in set $\mathcal{S}^{elem_1}$ and $\mathcal{S}^{elem_2}$ of ***Const.* 1** and **3**, respectively. Therefore, the candidate relations can be deduced for inclusions in set $\mathcal{S}^r = \{@, \geq\}$, and denoted as:

$$\{Time\} \times \{Econ\} \xrightarrow{cand.Rel} \{@, \geq\}$$

Finally, the correct relation of the paired elements, r=“@” will be predicted from the candidate set $\mathcal{S}^r = \{@, \geq\}$ by simultaneously satisfying the constraints on the given elements and the semantic descriptions of the input sentences using our proposed Transformer-based INN.

It should be noted that the proposed constraints can improve the performance of nested RE by reducing the number of relation types that need to be classified. Detailed experimental results are provided in the ablation experiment section.

4. Our SCL-nestedRE method

The overall architecture of our method called SCL-nestedRE for extracting nested relations is shown in Fig. 2, where it has

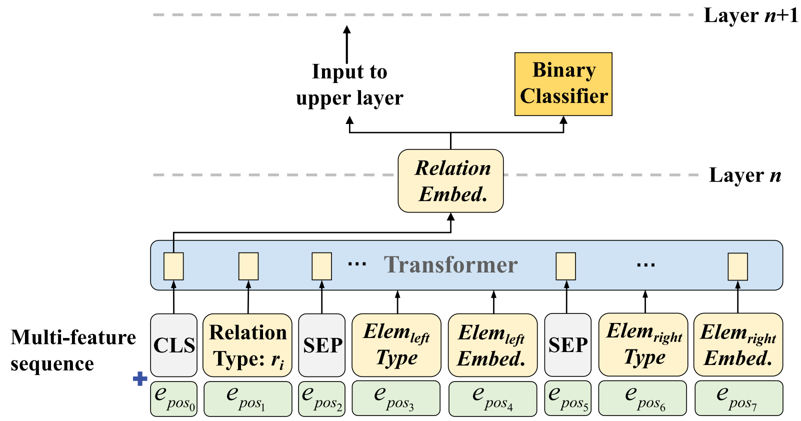


Fig. 3. Transformer-based iterative neural network model with multi-feature sequences as inputs for identification and extraction of relations.

three essential components: input sentence embedding layer, INN layers with multi-feature sequences for nested structure parsing, and SCL method guided by structure and semantic similarities for solving the low-data problem in nested RE.

4.1. Input embeddings

To represent the semantic descriptions of the input sentence, we use BERT (Devlin, Chang, Lee, & Toutanova, 2019) to encode the input character sequences and context information given that BERT has achieved great success in various NLP tasks. BERT is a type of Transformer model with a multi-layer structure and it provides the distributed semantic representations through two pre-training tasks: masked language modeling and next sentence prediction.

For the input word sequence of sentence $s = \{w_1, \ldots, w_i, \ldots, w_n\}$, we obtain the dense semantic vector of word w_i by summing up the corresponding word-piece and position embeddings after pre-training and fine tuning by BERT, denoted as:

$$e_{w_i} = e_{sem}(w_i) \tag{1}$$

where $e_{sem}(\cdot)$ represents the semantic lookup table, which is a parameter matrix and the ith vector in this matrix is the semantic embedding of the ith word in the given vocabulary.

4.2. INN with multi-feature sequences

After obtaining the semantic embeddings of the input words, we propose a Transformer-based iterative model for identifying and categorizing the nested relations in a layer by layer manner with the multi-feature sequences as the inputs for each layer. The overall structure of the input multi-feature sequence for the Transformer-based iterative model is shown in Fig. 3.

4.2.1. Multi-feature sequences

We convert the input relation triple, $<elem_l, RelType, elem_r>$, into a multi-feature sequence comprising four categories of feature embeddings: relation type embedding, element type embedding, special token embedding, and element embedding. The token "CLS" represents the beginning of the sequence and "SEP" is used to separate the relation type, element type, and element embeddings. The embeddings of the multiple features are described as follows.

Position embedding. To capture and represent the relative order information for the elements in the relations, the position embedding, e_{pos}, is added to the multiple feature embeddings and calculated as:

$$\begin{aligned} e_{pos_i} &= [e^0_{pos_i}, \ldots, e^t_{pos_i}, \ldots, e^{d-1}_{pos_i}]^T \\ e^t_{pos_i} &= \begin{cases} sin(i/10000^{2k/d}), \text{ if } t = 2k \\ cos(i/10000^{2k/d}), \text{ if } t = 2k+1 \end{cases} \end{aligned} \tag{2}$$

where e_{pos_i} is the position embedding for the ith input feature, $k \geq 0$ is an indicator, and d denotes the dimension of the position embedding.

Element type embedding. Considering that the elements comprise both entities and relations, we number the types of all the entities and relations, and then use the embedding layer to map each one-hot type number vector into an embedding vector:

$$e_{elemType} = e_{pos} + e_{numb}(elem) \tag{3}$$

where $e_{numb}(\cdot)$ represents the mapping function according to the number of element types.

Relation type embedding. Similar to element type embedding, the relation type embedding $e_{relType}$ is obtained by mapping the relation type number onto an embedding vector with an embedding layer:

$$e_{relType} = e_{pos} + e_{numb}(r) \tag{4}$$

Special token embedding. Two special tokens called "CLS" and "SEP" are introduced into the input sequence to represent the beginning token and separator token in the sequence, respectively. The embeddings of the special tokens are also obtained by mapping the token number onto an embedding vector:

$$\begin{aligned} e_{CLS} &= e_{pos} + e_{numb}(\text{“}CLS\text{”}) \\ e_{SEP} &= e_{pos} + e_{numb}(\text{“}SEP\text{”}) \end{aligned} \tag{5}$$

Element embedding. In the case where the element is an entity, the embedding is calculated for the hidden state of the element by using the attention mechanism (Mnih et al., 2014) over the input words included in the entity:

$$\begin{aligned} e_{elem='ent'} &= e_{pos} + \sum_{i=1}^{N} \alpha_i \cdot e_{w_i} \\ \alpha_i &= \frac{exp(s(e_{w_i}, q))}{\sum_{j=1}^{N}(s(e_{w_j}, q))} \\ s(e_{w_i}, q) &= \frac{e^T_{w_i} \cdot q}{\sqrt{D}} \end{aligned} \tag{6}$$

where e_{elem} denotes the hidden state embedding of the element, e_{w_i} represents the learned word embedding, D is the dimension of e_{w_i}, α_i is the weight of word w_i obtained using the attention mechanism, and $\boldsymbol{q}$ refers to the query vector, which is a randomly initialized trainable vector.

In the case where the element is a relation, the embedding of the element is represented by the embedding vector output from the feed-forward network in the Transformer model with the CLS token as the input.

$$e_{elem='rel'} = e_{pos} + W_1 \cdot Relu(W_2 \cdot e_{CLS} + b_1) + b_2 \tag{7}$$

where W^* and b^* are the trainable parameters defined in Transformer.

4.2.2. Nested RE with INN

Considering that new relation triples may be generated from the current layer and serve as the constituent elements of relations in the next layer, we iteratively extract the nested relation triples in a layer by layer manner in the multi-layer Transformer-based neural networks. For each layer, the nested RE procedure is conducted according to the following four steps.

Candidate relation triple generation. For the input sentence, we filter and obtain all possible relation triples by satisfying the constraints defined in the element pairs. First, we enumerate all of the element pairs denoted as $elemPair_{<i,j>} =< elem_l^i, elem_r^j >$ $(1 \le i \ne j \le |E \bigcup R|)$, where the subscripts i and j represent the ith and jth elements in the set $\{E \bigcup R\}$, respectively. The possible relation types set $\mathcal{S}^r = \{r_1, \ldots, r_k, \ldots, r_m | r_k \textit{ satisfies Const.k}, 1 \le k \le m\}$ can then be deduced according to Algorithm 1. Finally, all of the potential relation triples denoted as $r_{<i,j>,r_k} =< elem_l^i, r_k, elem_r^j >$, can be enumerated from the Cartesian product of $\mathcal{S}^{elemPair_{<i,j>}} \times \mathcal{S}^r$.

Multi-feature sequence construction. We further convert the candidate relation triples into the form of feature sequences. For one of the candidate relation triples that satisfy the constraints for the lth layer denoted as $r^l_{<i,j>,r_k} =< elem_l^i, r_k, elem_r^j >$, we convert $r^l_{<i,j>,r_k}$ into sequence form, $seq(r^l_{<i,j>,r_k}) = \{e_{CLS}, e_{relType='r_k'}, e_{SEP}, e_{elem_iType}, e_{elem_i}, e_{SEP}, e_{elem_jType}, e_{elem_j}\}$, as the input for Transformer for relation type prediction.

RE. We then categorize the relation types based on the multi-feature sequences derived from the element pairs. After using the multi-feature sequence $seq(r^l_{<i,j>,r_k})$ as the input for Transformer, a binary classifier is used to determine whether the relation type of $r^l_{<i,j>,r_k}$ is r_k or not with the loss function:

$$L_{relType} = -\frac{1}{M}\sum_{k=1}^{M} y_k log(p_k) + (1 - y_k) log(1 - p_k) \tag{8}$$

where y_k denotes the truth label, and $p_k = sigmoid(W_{relType} \cdot e_{elem='rel'} + b_{relType})$ represents the probability that the relation type of element pair $< elem_l^i, elem_r^j >$ is r_k, M is the number of possible relation types filtered by the constraints for the element pair, and W^* and b^* are the trainable parameters.

Element iteration. Finally, the positive relation triple with relation type r_k is set as the new element for nested RE in the next layer. If all the potential relation types $r_k (1 \le k \le m)$ for $elemPair_{<i,j>}$ are predicted as negative or the set of the potential relation type is empty after filtering by the constraints, we assume that there are no relations between the element pair. The iterative process continues until no new relation triples can be generated.

An example is shown in Fig. 2. Element pair <"2019", "*revenue*"> has two potential relation types comprising "@" and "≥" filtered by the constraints. Then, "@" is predicted as the positive relation by the classifier according to the semantic description. Finally, the extracted relation triple <"2019", "@", "*revenue*"> is set as the new element to form the nested relation triple <<"2019", "@", "*revenue*">, "−", <"2018", "@", "*revenue*">>. According to the constraints and extractor, there are no new relations for the element pairs in the third layer, so the process stops.

4.3. SCL-based optimization method

4.3.1. Motivation

Due to the special requirements of different application fields and the lack of annotated corpora, there is an urgent need to modify and optimize the existing nested RE method to adapt to low-data settings and reduce the cost of data annotation. To learn more distinguishable feature representations from limited labeled data, the contrastive learning method that is used widely in feature representation learning must learn the discriminability in the feature space of data by constructing positive and negative samples and by maximizing their semantic distance.

However, the positive and negative instances in nested RE are different from those in text classification, where the positive sample can be generated by using conventional data augmentation methods, such as synonym substitution, word deletion, reordering, or back-translation (Gao et al., 2021; Morris et al., 2020; Zhang et al., 2021). In the nested RE task, a sentence may contain multiple relations with nested structures and some of the relation triples may be similar or even identical to some parts of relation triples in another compared sentence, thereby behaving as the positive samples, whereas the other relation triples are different from the compared sentence, and thus they can serve as the negative samples. Therefore, given sentence i, even if we substitute the entities or relations with their synonyms, we still cannot ensure that the other sentences in the same batch will not contain some of the same or similar entities or relations compared with sentence i.

For example, in Fig. 4, sentence i and sentence j both have similar relation triples, such as <"2019", @, "*revenue*"> and <"2018", @, "*revenue*">, and different triples, such as <"−", ≥, "\$ 1 *million*"> in sentence i and <"/", ≥, "10%"> in sentence j.

Therefore, in nested RE, the partial and variable similarities between the two compared sentences determine how to generate positive and negative instance pairs, and measuring their similarities becomes particularly important but intractable.

According to the discussion above and the example given in Fig. 4, three phenomena can be found: **(1)** the similarity of the compared sentence pair can be subdivided into three situations: totally same, totally different, and partially the same and partially different; **(2)** the similarity between the compared two sentences is measured by the semantic similarity but it also involves the structure similarity, such as the number of layers and the same or different relation types in each layer, which have rarely been explored in previous studies; **(3)** when the structures are more similar, the semantics of the sentence pairs are more similar, whereas the semantics of the sentence pairs are less similar when their structures are less similar.

Therefore, we directly pair the input sentences and then perform contrastive learning based on these sentence pairs because the method involving manually or automatically generating positive or negative cases is not suitable for nested RE. We call this method **S**elf-**C**ontrastive **L**earning (**SCL**) because the sentence pairs are from the data set itself rather than external data or artificially generated. The proposed SCL method aims to obtain more representative feature representations guided by the structural and semantic similarities defined in the paired instances to boost the performance of the nested RE task.

4.3.2. SCL method

Based on the observations, we first pair the N input sentences in a batch and obtain the set of sentence pairs, $S^{pair} = \{(sen_i, sen_j) | 1 \le i < j \le N\}$, thereby extending the data scale to $N(N-1)/2$ sentence pairs. We then apply the SCL method based on the structure and semantic similarity to optimize the nested RE task.

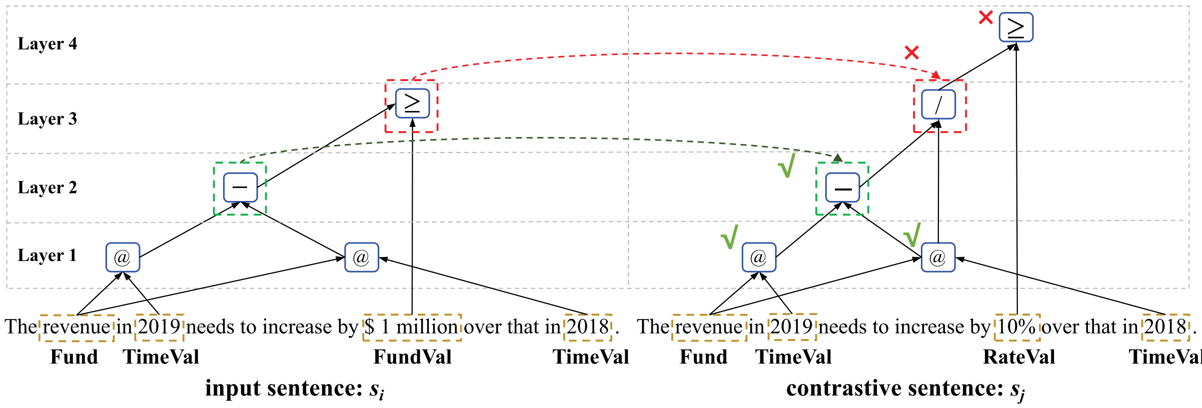


Fig. 4. Illustration of the proposed self-contrastive learning method guided by the structural and semantic similarity for the nested RE task. For the sentence pair (sen_i, sen_j), the two sentences share the same relation type and structure (green checkmark) in the lower two layers, but in the upper two layers, the relation types of the two sentences are different (red cross mark). Therefore, the two sentences are partially similar in terms of their structure and semantics.

Structure similarity. The structure similarity, denoted as $Sim_{struc}(sen_i, sen_j)$, between two paired sentences, denoted as $s^{pair}_{(i,j)} = (sen_i, sen_j)$, is defined by calculating the distribution of the same relation types in the nested structures of the two compared sentences.

Considering layer k as an example, we first count the number of the same relation r between sentence i and sentence j in layer k. Next, the numbers of all possible identical relations in layer $k+1$ are iteratively counted in the same manner. The statistical process ends when it reaches the highest layer of one of the two compared sentences. Finally, the structure similarity between the two compared sentences is obtained by calculating the ratio of the sum of the number of relations with the same type in each corresponding layer of the nested structure of sentence i and j relative to the total number of relations in the compared sentence pair.

The formula for $Sim_{struc}(sen_i, sen_j)$ is defined as follows:

$$sim_{struc}(sen_i, sen_j) = \frac{2}{M_1 + M_2} \cdot \sum_{k=1}^{min(K_i,K_j)} [\sum_{r \in L_i^k} min(c(L_i^k, r), c(L_j^k, r))] \tag{9}$$

where L_i^k represents the relation set of sentence i based on the kth layer and $c(L_i^k, r)$ is the number of relations with category r based on the kth layer, and M_i denotes the total number of relations in sentence i calculated as: $M_i = \sum_{k=1}^{K_i} |L_i^k|$, where K_i is the layer number of the nested structure in sentence i.

Semantic similarity. The semantic similarity, denoted as $sim_{sem}(sen_i, sen_j)$, is also used to evaluate the semantic similarity between the two compared sentences in sentence pair $s^{pair}_{(i,j)} \in \mathcal{S}^{pair}$, which is calculated as:

$$sim_{sem}(sen_i, sen_j) = \frac{exp(e(sen_i)^T \cdot e(sen_j))}{1 + exp(e(sen_i)^T \cdot e(sen_j))} \tag{10}$$

where $e(sen_i)$ represents the semantic embedding of sentence i calculated by the embedding of its beginning token "CLS", $e(\text{"}CLS\text{"})$. If the semantics of the two sentences are more similar, the value of $exp(e(sen_i)^T \cdot e(sen_j))$ is larger and the value of $sim_{sem}(sen_i, sen_j)$ tends to be 1. By contrast, when the two sentences are more semantically dissimilar, the value of $exp(e(sen_i)^T \cdot e(sen_j))$ is smaller and the value of $sim_{sem}(sen_i, sen_j)$ tends to be 0.

Self-contrastive loss for nested RE. To learn the feature representations from the input sentence pair by comparing the partial positive and partial negative parts in the given sentence pair simultaneously, a self-contrastive loss for sentence pair $s^{pair}_{(i,j)}$ is defined by minimizing the distance between the structure and semantic similarity of the sentence pair according to the observation that "*when the structures are more similar, the semantics of sentence pairs are more similar, and vice versa*". The loss function of SCL is calculated as:

$$L_{SCL} = \sqrt{\frac{2}{N(N-1)} \cdot \sum_{sen_i, sen_j \in \mathcal{S}^{pair}} (sim_{struc}(sen_i, sen_j) - sim_{sem}(sen_i, sen_j))^2} \tag{11}$$

where N represents the total sentence number.

4.4. Joint optimization for nested RE

After obtaining the contrastive loss with our proposed SCL method, the final optimization objective is joint training of the nested RE, defined in Eq. (8), and the SCL task is guided by the structure and semantic similarity, which is calculated as:

$$L = \lambda \cdot L_{SCL} + (1 - \lambda) \cdot L_{relType} \tag{12}$$

where λ is the hyper-parameter.

5. Experiments

5.1. Data sets and metrics

To empirically demonstrate that our method is not only competent for the nested RE task but also compatible with the flat RE tasks, we conducted experiments on three data sets.

Nested RE data set. Considering the almost total lack of public nested RE data sets, we generated a nested RE data set called **Policy** by annotating 3446 sentences in the collection of publicly issued Chinese preferential policies, where the acceptance requirements contain many nested relations between entities and relations. We defined 14 types of entities and 11 types of relations in the **Policy** data set. Descriptions of the defined entities and relations are given in Tables 1 and 2. We applied Algorithm 1 to the Policy data set and obtained the corresponding constraints for the left and right elements of the relations, as listed in Table 3. The Policy data set was split into training and validation sets at a ratio of 4:1, and 341 sentences were retained for testing. The detailed statistics are given in Fig. 5. The percentages of input sentences containing one, two, and ≥3 layers of relations were 43.79%, 33.95%, and 22.25%, respectively.

Flat RE data set. We used SemEval-2010 (Hendrickx et al., 2010) and KBP37 (Zhang & Wang, 2015) as the English flat RE

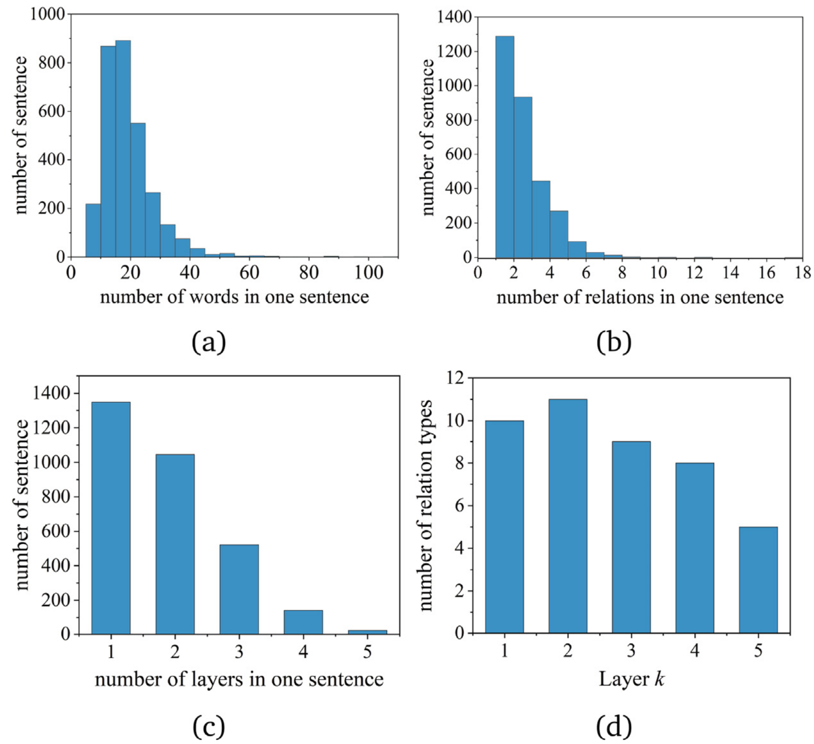


Fig. 5. Statistics of the Policy data set for nested RE task.

Table 1
Entities defined in the Policy data set.

Entity Type	Meaning
Org	Organization, institute, etc.
OrgQlf	Qualifications of organization.
Per	Personnel, shareholder, etc.
PerQlf	Qualifications of personnel.
Rt	Growth rate, interest rate, etc.
RtVal	Value of the rate
Econ	Economics, revenue, property, etc.
EconVal	Value of economics
Equip	Equipment, area, etc.
EquipVal	Value of equipment.
Time	Time, year, month, etc.
TimeVal	Value of time
Ser	Service, activity, product, etc
Addr	Address, location, etc.

Table 2
Relations defined in the Policy data set.

Relation type	Symbol
Greater than	$>$
Greater than or equal to	$\geq$
Less than	$<$
Less than or equal to	$\leq$
Equal to (Eql.)	$=$
Addition	$+$
Subtraction	$-$
Division	/
Attribution (Attr.)	@
Has	has
And	&&

data sets. Nine relations and one additional “other” relation are defined in SemEval-2010, and 18 types of relations are defined in KBP37, with one “other” relation. In SemEval-2010, 8000 instances were released for training and 2717 instances for testing. In KBP37, the number of sentences for training, validation, and testing were 15917, 1724, and 3405, respectively.

Most previous studies of RE used the precision (P), recall (R), and F1 score as assessment metrics. Precision (P) is the ratio between the number of correctly extracted relations and the total number of identified relations. Recall (R) is the ratio between the number of correctly extracted relations and the total number of gold relations. The F1 score is a harmonic average of the precision and recall to comprehensively evaluate the performance of the model. To ensure fair comparisons, we also obtained the average precision (P), recall (R), and F1 scores based on five runs of our method to evaluate the performance of our method and the compared baselines. The relation triples were regarded as correct only if all the constituent elements and the relative order between the elements were all correct.

5.2. Baseline methods

To demonstrate the superior performance and generality of our proposed SCL-nestedRE method, we selected several strong baseline methods for experimental comparisons under different RE tasks, i.e., nested RE and flat RE.

Baselines for nested RE. Considering that the nested RE task has rarely been studied previously, we only found one SOTA baseline method and several variants of the baseline to evaluate the effectiveness of our SCL-nestedRE method, as follows.

- **INN (LSTM+DAG-LSTM):** Cao et al. (2019) extracted nested relations with an INN, where the LSTM model was used to encode the input sentences and DAG-LSTM to parse the nested structure of the relations in a layer by layer manner.
- **BERT+DAG-LSTM:** This variant of the INN model was obtained by replacing the underlying LSTM model with BERT.
- **BERT+Transformer:** In the updated version of the INN model, we used the BERT model to replace the LSTM model in

Table 3
Constraints for left and right elements of the relations. The possible left and right element types of nested relations are listed in the second and third columns, respectively.

Relation type	Constraints for left elements	Constraints for right elements
$>$, $\geq$, $<$, $\leq$, $=$	*Econ, Rt, Per, Tim, Equip, @,* $+$, $-$, /	*Econ, EconVal, Rt, RtVal, Tim, TimVal, Equip, EquipVal, Per, @*
$+$, $-$, /	*Econ, Org, Per, Equip, Tim, Ser, @,* $+$, $-$	*Econ, Org, Per, Equip, Tim, Ser, @*
@	*Org, OrgQlf, Per, PerQlf, Tim, TimVal, Equip, Ser, Econ, Addr,* $>$, $\geq$, $<$, $\leq$, $=$, &&	*Org, Per, Ser, Tim, Equip, Econ, Rt, @*
has	*Org, Per, Ser, Equip, @*	*OrgQlf, Per, PerQlf, Ser, Equip*
&&	$>$, $\geq$, &&	$<$, $\leq$

Table 4
Main results for nested relation extraction based on Policy data set. The marker † refers to the statistical significance test *p-value* < 0.05 when compared between model SCL-nestedRE and other baseline methods, i.e., INN (Cao et al., 2019), BERT+DAG-LSTM, and BERT+Transformer.

Method		Precision	Recall	F1
From scratch	INN (LSTM+DAG-LSTM)	76.85 ± 2.28	68.52 ± 2.04	72.41 ± 0.82
	BERT+DAG-LSTM	84.23 ± 1.04	79.43 ± 2.03	81.74 ± 0.77
	BERT+Transformer (**ours**)	82.87 ± 2.37	81.94 ± 0.89	82.38 ± 0.98
	SCL-nestedRE (**ours**)	**85.48**† ± 1.80	**83.25**† ± 1.32	**84.33**† ± 0.84
Guided	INN (LSTM+DAG-LSTM)	83.98 ± 1.49	79.57 ± 1.87	81.70 ± 1.09
	BERT+DAG-LSTM	89.62 ± 0.88	87.67 ± 0.96	88.62 ± 0.46
	BERT+Transformer (**ours**)	88.44 ± 1.33	89.15 ± 0.72	88.78 ± 0.41
	SCL-nestedRE (**ours**)	**89.88**† ± 1.45	**89.98**† ± 1.29	**89.92**† ± 0.57

INN for semantic representation and the Transformer model to replace the DAG-LSTM model in INN for encoding the possible nested relations between entities and relations.

- **INN**$_{\textit{-guided}}$: In this variant of the original INN model, the subscript "***-guided***" indicates that the relations extracted in the current layer are replaced by the correct relations before predicting the relations in subsequent layers. The purpose of this design is to measure the impact of the error propagation from lower layers to higher layers.
- **BERT+DAG-LSTM**$_{\textit{-guided}}$: This is a variant of the BERT+DAG-LSTM model with the same modification "***-guided***".

Baselines for flat RE. To demonstrate that our SCL-nestedRE method could still work well in the flat RE task, several baseline methods designed for the flat RE task were selected for the experimental comparisons, as follows.

- **MTB:** This method proposed by Soares, FitzGerald, Ling, and Kwiatkowski (2019) uses the special token "BLANK" to replace entities in the relation statements for learning the relation representations by encouraging the relation representations to be similar if the "BLANK" token covers the same pairs of entities. BERT was adopted as the fundamental model for relation encoding.
- **LG-CNN:** Deng and Wu (2020) incorporated both the dependency encoded by the graph convolution network and the semantic information encoded by the convolutional neural network (CNN) into the LSTM model to capture the local context of the given sentences for RE.
- **TANL:** Paolini et al. (2021) proposed a generative model to cast the entity and relation extraction task as a translation task by designing augmented natural languages with specific patterns, and it has achieved SOTA performance.
- **KnowPrompt:** Chen et al. (2022) proposed a prompt tuning-based method for RE that injects the latent knowledge of relation labels into the constructed prompts. This method transforms the RE task into a masked language modeling task and it has achieved promising results.

In the following experiments, we re-implemented these baseline methods to the best of our ability.

5.3. Implementation details

The hyper-parameter and model settings for our method are described as follows. For the BERT model, we used the base version of BERT for comparative experiments where the dimension of the hidden states for the input characters was set to 768. For the Transformer model, the dimensions of the hidden states for the input relation elements and their corresponding types were all set to 768. The model was trained with Adam (Kingma & Ba, 2015) and the learning rate was set to 1e−5 for all data sets. We set the dropout rate to 0.1 for all data sets. The batch sizes were set to 8 and 16 for the nested RE and flat RE data sets, respectively. Our proposed model was trained for 40 epochs and stopped when the performance did not improve for 20 epochs. The maximum sentence length was set to 128 for the nested RE data set and 512 for the flat RE data sets. All of our experiments were conducted with one NVIDIA RTX-3090 GPU with 24G of graphical memory. For the flat RE data sets, we followed the same training/validation/test data split as the baseline methods (Hendrickx et al., 2010; Zeng, Zeng, He, Liu, & Zhao, 2018; Zhang & Wang, 2015).

In the following experiments, we re-implemented the baseline methods to the best of our ability according to the original studies. We used the paired t-test to assess statistically significant differences, where the test was one-tailed. In each round of testing, we used the same random seed to initialize our proposed model and the baseline model, and then changed the random seed and repeated the experiments five times. Finally, we conducted the paired t-test for all of the result pairs.

6. Results and analyses

6.1. Effectiveness analysis

Main results with different baselines. As shown in Table 4, the experiments for nested RE with different baselines were performed under two settings: "**From scratch**" and "**Guided**". "From scratch" indicates that the models extracted the relations in a layer by layer manner, but the incorrectly identified relations were also used as the inputs to predict the relations in the

Table 5
Main results for different layers in nested RE with the Policy data set. F1 scores are reported. The marker † refers to the statistical significance test *p-value* < 0.05 when compared between model SCL-nestedRE and other baseline methods, i.e., INN (Cao et al., 2019), BERT+DAG-LSTM, and BERT+Transformer.

Method		Layer $L = 1$	$L = 2$	$L \geq 3$
From scratch	INN (LSTM+DAG-LSTM)	80.51 ± 0.91	66.85 ± 1.89	38.69 ± 4.51
	BERT+DAG-LSTM	87.39 ± 0.74	78.24 ± 1.18	57.92 ± 2.81
	BERT+Transformer (**ours**)	88.54 ± 0.98	79.12 ± 0.95	57.27 ± 1.77
	SCL-nestedRE (**ours**)	**89.60**† ± 0.62	**81.40**† ± 1.36	**62.70**† ± 3.50
Guided	INN (LSTM+DAG-LSTM)	80.51 ± 0.91	81.66 ± 1.74	88.39 ± 1.14
	BERT+DAG-LSTM	87.39 ± 0.74	90.25 ± 0.73	**90.31**† ± 2.75
	BERT+Transformer (**ours**)	88.54 ± 0.98	88.99 ± 1.33	89.36 ± 2.06
	SCL-nestedRE (**ours**)	**89.60**† ± 0.62	**90.31**† ± 0.97	89.72 ± 2.45

Table 6
Ablation experimental results with the Policy data set. (From scratch.)

Method variants	Precision	Δ	Recall	Δ	F1	Δ
SCL-nestedRE (ours)	**85.48**	Δ	**83.25**	Δ	**84.33**	Δ
-w/o pos	81.95	−3.53	71.53	−11.72	76.33	−8.00
-w/o elemType	83.41	−2.07	78.39	−4.86	80.78	−3.55
-w/o pos + elemType	80.62	−4.86	71.36	−11.89	75.67	−8.66
-w/o constraints	80.28	−5.20	82.98	−0.27	81.57	−2.76
-w/o SCL (i.e., BERT+Transf)	82.87	−2.61	81.94	−1.31	82.38	−1.95
-w learnable pos	84.05	−1.43	81.49	−1.76	82.73	−1.60

subsequent layers, thereby leading to error propagation. "Guided" indicates that the relations extracted in the current layer were replaced by the correct relation triples before predicting the relations in the next layer. The "Guided" setting was introduced to evaluate the effect of error propagation as well as the layers.

Using the "From scratch" setting of our proposed full version method, SCL-nestedRE outperformed the SOTA method comprising INN (Cao et al., 2019) by 11.92 in terms of the F1 score, thereby demonstrating the effectiveness and superior performance of our method. We also compared the performance with different variants of the baseline model. After replacing the underlying LSTM model with BERT, the F1 score increased by 9.33 compared with the basic INN method. Moreover, the performance was improved further by 0.64 in terms of the F1 score when we replaced the upper DAG-LSTM model for relation categorization with Transformer. The huge improvement in performance validates the great power of the prevailing pre-trained language model, i.e., BERT, for semantic representation because it plays a crucial role in nested RE, and this also explains why we selected BERT as the backbone of our method.

Using the "Guided" setting, the overall performance of the listed methods was better than the corresponding methods in the "From scratch" setting, which illustrated that the error propagation does reduce the performance of nested relation extraction. Similarly, our method achieved the SOTA results, surpassing method INN by 8.22 in terms of the F1 score. The BERT and Transformer-based variants also outperformed the basic LSTM-based model, which is consistent with the phenomenon in the "From scratch" setting. Considering most of the previous studies selected precision (P), recall (R), and F1 score as the metrics, we used these three metrics in the follow-up experiments for fair comparisons.

Main results for different layers. According to the experimental results listed in Table 5, three interesting outcomes were found after further investigating the error propagation effect along the layers of the nested structure of relations.

(1) When the number of relation layers increased, the recognition performance of all compared methods decreased in the "From scratch" settings. For example, when the number of layers increased from L=1 to $L \geq 3$, the F1 score using the INN method dropped from 80.51 to 38.69. The possible explanation is that if a correct relation in the lower layer is not extracted, the relation triples based on it in the higher layer will not be correctly recognized. In addition, if an incorrect relation is extracted from the lower layer, a series of relation triples may be falsely generated in the higher layer based on that incorrect relation. Therefore, it is a non-trivial task to extract nested relations from the unstructured text.

(2) However, with the "Guided" setting, the F1 score of the compared methods increased as the layers rise, which indicates that replacing the predicted relations at each layer with the correct relation triples can greatly fix the error propagation problem along with the layers. Another possible reason for performance improvement is that: According to the statistics in Fig. 5, the number of relation types in the higher layer is less than that in the lower layer. For example, in the third, fourth, and fifth layers, there are only 9, 8, and 5 relation types left, respectively, out of a total of 11 pre-defined relation types. Meanwhile, the size of the training data set corresponding to each relation type in the higher layer is also increased through the guided-based correction strategy. Therefore, the performance of nested relation extraction methods improved because of the reduced number of relation categories and the expanded scale of the training data set.

(3) Whether in the setting of "From scratch" or "Guided", our method surpassed the baseline methods across the multiple layers of the nested relations in most cases, which verifies the remarkable adaptability and effectiveness of our proposed method.

The detailed analyses of the contributions from different components of our method were given in the following ablation experiments.

6.2. Ablation analysis

To quantitatively evaluate the contributions of different components of our proposed method with the multi-feature sequences and SCL method, several ablation experiments were conducted and the results are listed in Table 6.

Table 7
Ablation experiments for constraints between relation elements on the Policy data set, using the "From scratch" setting. Each method was conducted five times for statistical significance testing. F1 values are reported. "-w constraints" means that the constraints between elements were applied to the compared methods, while "-w/o constraints" means that the constraints were removed. AVG means the averaged value of the five time experiments.

Methods	-*w* constraints						-*w/o* constraints					
	1st	2nd	3rd	4th	5th	AVG	1st	2nd	3rd	4th	5th	AVG
INN (LSTM+DAG-LSTM)	72.16	73.59	71.83	72.87	71.58	72.41	65.29	67.90	66.77	67.48	66.11	66.71
BERT+DAG-LSTM	81.28	80.68	82.53	81.82	82.39	81.74	77.86	80.06	78.57	77.73	78.49	78.54
BERT+Transformer (**ours**)	83.08	83.04	81.80	83.06	80.91	82.38	80.82	82.26	79.88	78.94	82.88	80.96
SCL-nestedRE (**ours**)	83.71	83.53	84.44	85.67	84.31	84.33	81.57	81.28	82.18	79.17	83.65	81.57

Table 8
Experimental results with flat relation extraction.

Data set	KBP37			SemEval-2010		
Methods	Precision	Recall	F1	Precision	Recall	F1
MTB (Soares et al., 2019)	–	–	69.30	–	–	89.50
LG-CNN (Deng & Wu, 2020)	–	–	63.20	–	–	85.50
TANL (Paolini et al., 2021)	67.33	71.80	**69.50**	87.85	91.07	89.43
KnowPrompt (Chen et al., 2022)	–	–	**69.50**	–	–	**89.59**
INN (LSTM+DAG-LSTM) (Cao et al., 2019)	45.67 ± 0.98	47.24 ± 1.10	46.43 ± 0.70	64.43 ± 1.69	64.98 ± 2.74	64.65 ± 0.88
BERT+DAG-LSTM	68.47 ± 0.85	64.51 ± 0.69	66.43 ± 0.73	84.35 ± 0.94	83.00 ± 0.74	83.67 ± 0.61
BERT+Transformer (**ours**)	69.06 ± 0.94	65.26 ± 0.90	67.10 ± 0.58	84.82 ± 0.60	84.06 ± 0.55	84.43 ± 0.24

After removing the position embedding feature denoted as "-*w/o pos*", the F1 score with our method decreased greatly by 8 from 84.33 to 76.33. This result shows that the position information between the elements of the relations is essential for identifying the correct order of the elements in the relation triples. To further evaluate the effects of different positional embeddings on the model''s performance, we also used the positional embedding in a learnable manner as the variant method, which is denoted as "-*w* learnable *pos*" in the last line of Table 6. We found that after applying learnable positional embedding, the variant method "-*w* learnable *pos*" performed worse than the original method by 1.60 in terms of F1.

Removing the embedding features for the element type denoted as "-*w/o elemType*" reduced the performance in terms of all evaluation metrics, i.e., from 85.48 to 83.41 in terms of the precision, from 83.25 to 78.39 in terms of the recall, and from 84.33 to 80.78 in terms of the F1 score. Intuitively, it can be assumed that the explicit element type embedding plays as a role of the hint, which informs the model that some types of input element pairs may tend to form specific types of relational triples.

Moreover, removing both the position and element type embedding denoted as "-*w/o pos + elemType*" further reduced the performance of the proposed SCL-nestedRE method, i.e., from 84.33 to 75.67, which highlights the superposition effect of our introduced multiple features on the performance of our method.

After removing the constraints between elements denoted as "-*w/o constraints*", the performance of our method also decreased by 2.76 in terms of the F1 score, which indicates the reduction of search space can improve the performance of nested relation extraction. To better verify the necessity of these constraints, we designed ablation experiments through further removing these constraints from other baseline methods. The experimental results are shown in Table 7.

According to the results, we found that when these constraints are removed, not only the performance of our method decreased, but also the performance of other baseline methods, thereby demonstrating that the proposed constraints between relation elements are effective and universal for improving the performance of the nested RE task.

In addition, we also performed paired statistical significance testings based on the experimental results listed in Table 7 to determine whether the proposed constraints between relation elements can significantly improve the performance of the nested RE task.

For example, for method SCL-nestedRE$_{-w\ constraints}$, the results of five time experiments are {83.71, 83.53, 84.44, 85.67, 84.31}, respectively. For the ablated method SCL-nestedRE$_{-w/o\ constraints}$, the results of five time experiments are {81.57, 81.28, 82.18, 79.17, 83.65}, respectively. According to the paired significance testing results, the p-value < 0.05. Therefore, the constraints proposed above are effective for improving the performance of nested RE.

In addition, our presented SCL method had a positive role in boosting the performance of the nested RE task. After removing the SCL denoted as "-*w/o SCL*", the performance decreased in terms of all the metrics, i.e., from 85.48 to 82.87 in terms of the precision, from 83.25 to 81.94 in terms of the recall, and from 84.33 to 82.38 in terms of the F1 score.

6.3. Generality analysis

Considering that the nested RE is a more general structure compared with the flat RE, we conducted the following experiments to prove that our proposed SCL-nestedRE method can also deal with the flat relation extraction task with more superior or competitive performance.

Performance with flat RE. As shown in Table 8, the MTB, LG-CNN, TANL, and KnowPrompt methods are the SOTA baselines designed specifically for the flat RE task. There is no nested structure in the plane relation, so we removed the self-contrastive loss component from our full version model and obtained a variant of our method denoted as BERT+Transformer. With the KBP37 data set, the TANL and KnowPrompt methods achieved the best performance, where the F1 scores were 0.2 higher than that for the second best method comprising MTB. Our BERT+Transformer method outperformed the LG-CNN method by 3.9 in terms of the F1 score and it achieved competitive performance compared to the SOTA methods comprising MTB, TANL, and KnowPrompt, thereby demonstrating that our nested RE method can still perform well for relations in a flat structure. With the SemEval-2010 data set, our method did not perform as well as the best method KnowPrompt, but it still outperformed the LG-CNN method, thereby showing the effectiveness and generality of our method for the flat RE task. Another interesting observation is that despite some performance degeneration, the INN method, designed for the nested RE task, was also compatible with the flat RE task. In addition, after replacing the underlying LSTM network in INN with BERT, the F1 scores improved with the

Table 9
Verification of the portability of the self-contrastive loss based on the Policy data set for the nested RE task. "-*w* self-contrastive loss (*SCL*)" means that the self-contrastive loss is added to the baseline methods, i.e., INN (Cao et al., 2019) and BERT+DAG-LSTM.

Method variant	Precision	Recall	F1
INN (LSTM+DAG-LSTM)	**76.85**	68.52	**72.41**
-*w* self-contrastive loss	74.27	**68.73**	71.34
BERT+DAG-LSTM	84.23	**79.43**	81.74
-*w* self-contrastive loss	**85.37**	79.26	**82.19**

Table 10
Experimental results under different training data size percentages based on the Policy data set. The marker † refers to the statistical significance test *p-value* < 0.05 when compared between the SCL-nestedRE model and other baseline methods, i.e., INN (Cao et al., 2019), BERT+DAG-LSTM, and BERT+Transformer.

Data size	Methods	Precision	Recall	F1
10%	INN (LSTM+DAG-LSTM)	61.58 ± 1.57	48.64 ± 2.21	54.30 ± 0.87
	BERT+DAG-LSTM	64.25 ± 3.43	58.54 ± 1.82	61.23 ± 2.08
	BERT+Transformer (**ours**)	66.59 ± 4.28	**63.96**† ± 3.03	65.12 ± 1.47
	SCL-nestedRE (**ours**)	**68.20**† ± 1.72	63.34 ± 1.95	**65.65**† ± 1.07
20%	INN (LSTM+DAG-LSTM)	61.00 ± 2.27	60.57 ± 3.95	60.67 ± 1.56
	BERT+DAG-LSTM	71.90 ± 1.74	64.41 ± 2.21	67.90 ± 0.63
	BERT+Transformer (**ours**)	**72.81**† ± 2.42	69.36 ± 3.28	70.96 ± 1.03
	SCL-nestedRE (**ours**)	72.54 ± 2.35	**70.97**† ± 3.11	**71.67**† ± 0.84
30%	INN (LSTM+DAG-LSTM)	68.35 ± 2.40	58.12 ± 2.56	62.75 ± 0.89
	BERT+DAG-LSTM	76.01 ± 3.94	70.49 ± 1.77	73.08 ± 1.44
	BERT+Transformer (**ours**)	72.54 ± 4.81	**74.72**† ± 2.56	73.57 ± 3.31
	SCL-nestedRE (**ours**)	**76.60**† ± 1.85	74.16 ± 1.62	**75.34**† ± 0.91
50%	INN (LSTM+DAG-LSTM)	67.62 ± 2.12	64.41 ± 1.77	65.94 ± 0.69
	BERT+DAG-LSTM	79.98 ± 0.63	76.16 ± 2.13	78.00 ± 1.06
	BERT+Transformer (**ours**)	76.57 ± 1.95	**79.76**† ± 2.28	79.63 ± 0.70
	SCL-nestedRE (**ours**)	**81.76**† ± 0.44	79.14 ± 1.46	**80.43**† ± 0.96
100%	INN (LSTM+DAG-LSTM)	76.85 ± 2.28	68.52 ± 2.04	72.41 ± 0.82
	BERT+DAG-LSTM	84.23 ± 1.04	79.43 ± 2.03	81.74 ± 0.77
	BERT+Transformer (**ours**)	82.87 ± 2.37	81.94 ± 0.89	82.38 ± 0.98
	SCL-nestedRE (**ours**)	**85.48**† ± 1.80	**83.25**† ± 1.32	**84.33**† ± 0.84

two data sets, thereby again demonstrating the great power of the pre-trained language model. These experimental results may inspire future studies to develop a universal model for both flat and nested RE tasks.

Portability of SCL method. To evaluate the portability of our proposed SCL method, we transferred the self-contrastive loss into the existing INN method and our BERT+DAG-LSTM method and conducted experiments with the Policy data set. According to the experimental results shown in Table 9, after incorporating the SCL strategy, the performance of INN+self-contrastive loss decreased in terms of the F1 score compared with the vanilla INN method, and the BERT+DAG-LSTM method with the self-contrastive loss outperformed the method without the self-contrastive loss by 0.45 in terms of the F1 score. These results may be explained by the INN method utilizing the average value of the input word embeddings encoded by the LSTM model to represent the semantic embeddings of sentences, whereas the BERT+DAG-LSTM method used the embedding of the special token "CLS" provided by the BERT model to represent the semantic embeddings of sentences, which could obtain higher quality semantic representations for input sentences. The experimental results showed that under the condition of the availability of high-quality semantic representations for sentences, our SCL strategy could further improve the performance of existing methods without changing their overall structures. The improvements in the experimental results also indicate that introducing the nested structure information is beneficial for the nested RE task.

6.4. Studies in low-data settings

To verify the effectiveness of our proposed method in low-data settings, which are quite common in practical application scenarios, we randomly extracted 10%, 20%, 30%, and 50% of the original Policy training data set to generate three new subsets of training data. We trained and evaluated the performance of our SCL-nestedRE method and the baseline methods based on these training data subsets and the experimental results are given in Table 10. According to the experimental results, we found the following.

(1) Our SCL-nestedRE method outperformed the other three baseline methods with all data sets of different sizes, thereby demonstrating that our method is highly suitable for low-data settings and more reliable when the scale of the annotated data is limited.

(2) The performance measured using the precision, recall, and F1 score for all of the methods listed in Table 9 increased as the scale of the data set increased from 10% to 100%, thereby confirming that the model performed better when the data set was larger.

(3) Compared with BERT+Transf method, the variant of SCL-nestedRE obtained by removing the self-contrastive loss, the performance of the full version SCL-nestedRE method was superior to the simplified version, which indicates that the SCL-based optimization strategy plays a crucial role in solving the low-data problem by fully exploiting the nested structure information between the input sentence pairs. In summary, our proposed method is highly practical and it can work well in low-data settings.

Considering that the data distribution of each relation type in the Policy data set is uneven, we conducted the N-way, K-shot experiments based on the five lowest frequency relations, (i.e., "$<$", "$\leq$", "$=$", "/" and "and") among all of the 11 relation types defined in the Policy data set to evaluate the performance of the nested RE methods in few-shot settings. According to the

Table 11
Average experimental results for low-frequency relations in few-shot settings based on the Policy data set. The selected five lowest frequency relations are relations "<", "≤", "=", "/" and "&&". The marker † refers to the statistical significance test *p-value* < 0.05 when compared between model SCL-nestedRE and other baseline methods, i.e., INN (Cao et al., 2019), BERT+DAG-LSTM, and BERT+Transformer.

N-way, K-shot	Methods	Precision	Recall	F1
5-way, 1-shot	INN (LSTM+DAG-LSTM)	9.23 ± 9.36	7.03 ± 3.33	6.02 ± 4.59
	BERT+DAG-LSTM	5.09 ± 3.31	30.27 ± 24.26	7.39 ± 3.58
	BERT+Transformer (**ours**)	**22.23**† ± 5.71	32.30 ± 15.99	24.33 ± 3.78
	SCL-nestedRE (**ours**)	20.21 ± 2.96	**43.24**† ± 8.40	**27.05**† ± 1.23
5-way, 5-shot	INN (LSTM+DAG-LSTM)	45.85 ± 4.68	16.22 ± 1.79	23.87 ± 2.01
	BERT+DAG-LSTM	**56.02**† ± 24.73	17.03 ± 4.52	24.28 ± 3.15
	BERT+Transformer (**ours**)	39.25 ± 13.55	21.89 ± 4.84	27.52 ± 5.83
	SCL-nestedRE (**ours**)	35.39 ± 5.43	**29.46**† ± 6.13	**31.49**† ± 4.11
5-way, 10-shot	INN (LSTM+DAG-LSTM)	43.21 ± 25.44	20.41 ± 10.54	26.46 ± 12.28
	BERT+DAG-LSTM	**56.24**† ± 10.59	18.92 ± 6.89	27.15 ± 5.09
	BERT+Transformer (**ours**)	48.48 ± 10.51	25.81 ± 6.38	32.59 ± 3.11
	SCL-nestedRE (**ours**)	36.66 ± 3.56	**38.11**† ± 9.98	**36.55**† ± 4.45

few-shot experimental results listed in Table 11, we obtained the following findings.

(1) The large reduction in the number of samples under each relation type greatly reduced the performance of the compared methods, but the performance of our method still exceeded that of the baseline methods, thereby verifying the effectiveness of our method in dealing with few-shot settings.
(2) In addition, to demonstrate the contribution of the nested structure information formalized by the self-contrastive loss, we compared the results obtained by our SCL-nestedRE method and the BERT+Transformer method, where the self-contrastive loss was removed. SCL-nestedRE outperformed BERT+Transformer by 2.72, 3.97, and 3.96 in terms of the F1 score in settings with 1, 5, and 10 shots, respectively.
(3) According to the experimental results in Table 11, we also found that the BERT-based methods, i.e., BERT+DAG-LSTM and BERT+Transformer, were superior to the LSTM-based method, i.e., INN, which are similar to the results shown in Table 10. The backbone model plays a crucial role in semantic representation for the RE task.

In summary, according to the experimental results obtained in low-data settings listed in Tables 10 and 11, we can make the following two conclusions.

(1) The introduction of the nested structure-based optimization strategy, i.e., the proposed self-contrastive loss, even boosted the performance of the nested RE task, in low-data settings, which have been rarely studied previously.
(2) Replacing the underlying neural network, such as LSTM, in the baseline model with a more powerful pre-training language model, such as BERT and Transformer, improved the performance in the nested RE task.

6.5. Effects of hyper-parameters

To evaluate the contribution of the self-contrastive loss, we conducted experiments to assess the performance of our proposed SCL-nestedRE method when adjusting the weight parameter λ of the self-contrastive loss, and the results are given in Fig. 6. According to Fig. 6a, for the relations in all layers (*Layer L=all*), the performance (measured by the F1 value) of our method improved initially as the parameter λ increased. When $\lambda = 0.1$, the F1 value reached the maximum value, but the F1 value then decreased as the parameter λ increased further.

Table 12
Comparison between different backbones in the nested relation extraction task with the "From scratch" setting. The marker † refers to the statistical significance test *p-value* < 0.05 when compared between the BERT-based and RoBERTa-based models.

Task type	Nested RE		
Data set	Policy		
Method variants	Precision	Recall	F1
SCL-nestedRE$_{BERT}$ (**ours**)	85.48 ± 1.80	**83.25**† ± 1.32	84.33 ± 0.84
SCL-nestedRE$_{RoBERTa}$ (**ours**)	**87.31**† ± 1.12	83.10† ± 1.78	**85.14**† ± 0.72

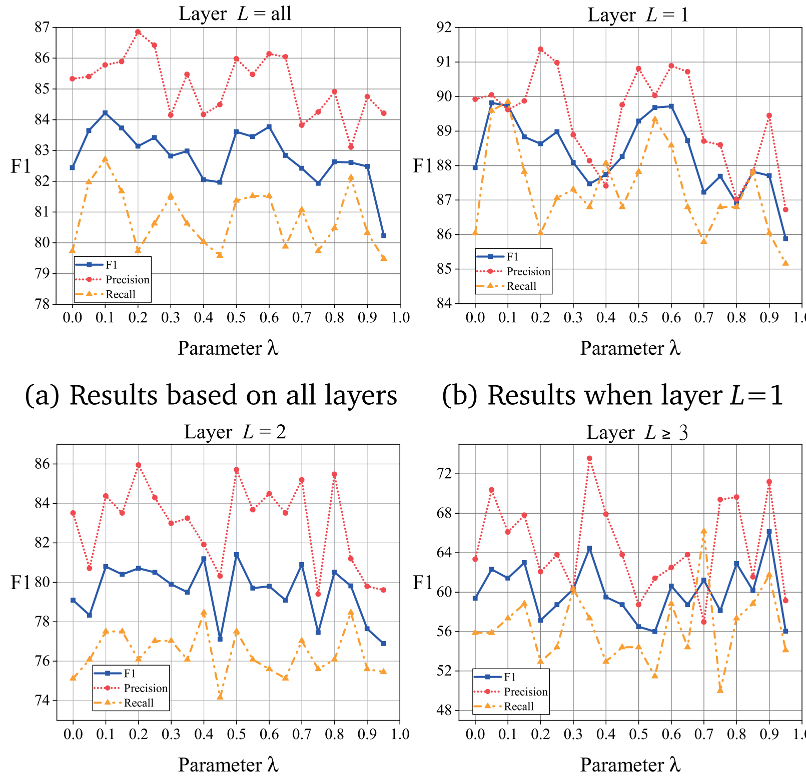
We also statistically analyzed the recognition performance for nested relations in different layers (layer $L = 1$, $L = 2$, and $L \geq 3$), and the results are shown in Fig. 6b, 6c, and 6d, respectively. The results showed the following. **(1)** The performance improved initially and then declined as the parameter λ increased, which is similar to the trend shown in Fig. 6a. **(2)** For relations in the lower layers, layer $L = 1$ and 2, when the parameter λ was set to 0.05 and 0.5, our SCL-nestedRE method reached the best performance, respectively. For relations in the higher layers, layer $L \geq 3$, our method performed best when the parameter λ increased further to 0.9.

It is possible that when the number of layers where relations were located increased, the hierarchical structure information for the nested relationship began to play an important role in improving the performance of nested RE. Therefore, for relations in higher layers, when the weight parameter λ corresponding to the self-contrast loss increased to an appropriate amount, the self-contrastive loss comprising the structure and semantic similarity was more helpful for improving the performance of our method.

6.6. Compatibility analysis

Considering that the BERT model has unprecedented power for representation learning in the NLP field and it contains abundant latent semantic knowledge, we used BERT as the backbone for our method.

Furthermore, to verify the compatibility of our method in terms of the design structure, we replaced BERT with the RoBERTa model of the base version as the backbone. As an evolutionary version of BERT, RoBERTa is equipped with the dynamic mask mechanism and byte-pair encoding, and it has achieved great success in many application fields. The experimental results are shown in Tables 12 and 13.



(a) Results based on all layers (b) Results when layer $L=1$

(c) Results when layer $L=2$ (d) Results when layer $L \geq 3$

Fig. 6. Effects of hyper-parameter λ in SCL-nestedRE method with the Policy data set when layer L varied.

Table 13
Comparison between different backbones in the flat relation extraction task.The marker † refers to the statistical significance test *p-value* $<$ 0.05 when compared to the BERT-based model. "Transf" is the abbreviation for "Transformer".

Task type	Flat RE					
Data set	KBP37			SemEval-2010		
Method variants	Precision	Recall	F1	Precision	Recall	F1
BERT+Transf (**ours**)	69.06 ± 0.94	65.26 ± 0.90	67.10 ± 0.58	84.82 ± 0.60	84.06 ± 0.55	84.43 ± 0.24
RoBERTa+Transf (**ours**)	**70.13**† ± 0.74	**66.02**† ± 1.48	**68.00**† ± 0.65	**87.16**† ± 1.03	**84.43**† ± 0.96	**85.77**† ± 0.51

According to Tables 12 and 13, we obtained the following results. (1) In the nested RE task based on the Policy data set, the performance of our variant method SCL-nestedRE$_{RoBERTa}$ increased from 84.33 to 85.12 in terms of the F1 score after replacing BERT with RoBERTa, and its performance improved further compared with other baseline methods. (2) In the flat RE task, RoBERTa+Transf still outperformed BERT+Transf by 0.9 and 1.34 in terms of the F1 scores with the KBP37 and SemEval-2010 data sets, respectively. These experimental results demonstrate the compatibility of our proposed approach because our model was compatible with different backbones, i.e., the BERT-based backbone and the RoBERTa-based backbone, and the overall performance of our variant model was better when the backbone model was better.

6.7. Sensitivity analysis

To evaluate the sensitivity of our method to entity identification errors, we modified the proposed method into a pipelined form. The pipelined method first extracted the entities from the input sentences and then extracted the nested relations. We used the LSTM+CRF and BERT + CRF models for named entity recognition. In Table 14, the variant nested RE model using LSTM+CRF for entity recognition is denoted as SCL-nestedRE$_{LSTM+CRF}$ and the variant nested RE model using BERT+CRF for entity recognition is denoted as SCL-nestedRE$_{BERT+CRF}$. The nested RE model given the golden result for entity recognition is denoted as SCL-nestedRE$_{Gold}$. In addition, the experimental standard deviation and statistical significance test results for named entity recognition and nested

Table 14
Comparison of the sensitivity of entity recognition errors to our nested relation extraction method. The marker † refers to the statistical significance test *p-value* $<$ 0.05 when compared to LSTM+CRF model.

Data set	Policy		
Method	Precision	Recall	F1
SCL-nestedRE$_{Gold}$	**85.48**† ± 1.80	**83.25**† ± 1.32	**84.33**† ± 0.84
SCL-nestedRE$_{LSTM+CRF}$	64.66 ± 1.46	49.78 ± 0.77	56.24 ± 0.64
SCL-nestedRE$_{BERT+CRF}$	64.51 ± 2.05	57.32 ± 0.76	60.68 ± 0.60

Table 15
Experimental results for named entity recognition based on the Policy data set. The marker † refers to the statistical significance test *p-value* < 0.05 when compared to LSTM+CRF model.

Data set	Policy		
Method	Precision	Recall	F1
LSTM+CRF	86.06 ± 0.62	78.67 ± 0.22	82.20 ± 0.40
BERT+CRF	**86.60**† ± 0.40	**84.33**† ± 0.99	**85.45**† ± 0.34

Table 16
Experiment results in terms of model efficiency between the models compared comprising INN (Cao et al., 2019), BERT+DAG-LSTM, SCL-nestedRE, SCL-nestedRE$_{RoBERTa}$, BERT+Transformer, and RoBERTa+Transformer with three data sets: Policy, KBP37, and SemEval-20210. *Sent./s* represents the number of sentences that could be processed per second by each model. The string "↑" after "*Sent./s*" shows that the result was better when the indicator was higher, i.e., the computational efficiency of the model was higher. SCL-nestedRE$_{RoBERTa}$ is a variant model obtained by replacing BERT with RoBERTa.

Task type	Data set	Method	*Sent./s*↑	F1
Nested RE	Policy	INN(LSTM+DAG-LSTM)	149.56	72.41
		BERT+DAG-LSTM	78.68	81.74
		SCL-nestedRE (**ours**)	58.35	84.33
		SCL-nestedRE$_{RoBERTa}$(**ours**)	40.65	**85.14**
Flat RE	KBP37	INN(LSTM+DAG-LSTM)	183.40	46.43
		BERT+DAG-LSTM	85.41	66.43
		BERT+Transformer (**ours**)	78.75	67.10
		RoBERTa+Transformer (**ours**)	60.68	**68.00**
	SemEval-2010	INN(LSTM+DAG-LSTM)	294.05	64.65
		BERT+DAG-LSTM	98.49	83.67
		BERT+Transformer (**ours**)	89.46	84.43
		RoBERTa+Transformer (**ours**)	64.42	**85.77**

RE are reported in Tables 14 and 15. The experimental results are listed in Tables 14 and 15.

According to Tables 14 and 15, the results showed the following.

(1) The performance of the entity recognition method based on BERT+CRF was higher than that based on LSTM+CRF with the Policy data set, where the F1 value increased by 3.25. This result also shows that the pre-trained language model BERT has strong semantic representation power.
(2) When the entity recognition performance was higher, the performance of the nested RE model based on entity recognition was also higher. The method SCL-nestedRE$_{BERT+CRF}$ consistently outperformed SCL-nestedRE$_{LSTM+CRF}$ by 4.44 in terms of the F1 score. However, both the pipelined methods performed worse than the original method with the golden entities, i.e., SCL-nestedRE$_{Gold}$, in terms of the F1 score, thereby demonstrating that the performance of the nested RE task depended on the quality of the recognized entities in the bottom layer.

The results obtained in these two experiments demonstrate that the entity recognition performance played a positive role in improving the performance of nested RE.

6.8. Computational efficiency

To evaluate the computational efficiency of our proposed approach and the baseline methods, we conducted model efficiency assessment experiments based on the number of input sentences per second processed by the models with the Policy, KBP37, and SemEval-2010 data sets during the inference process. The experimental results are presented in Table 16.

The results showed the following. (1) The computational efficiency of LSTM-based methods, i.e., INN, was higher than that of BERT-based methods and RoBERTa-based methods, such as BERT+DAG-LSTM and our methods, mainly due to the complexity of the backbone model. However, the RoBERTa-based methods obtained higher F1 values for nested RE, thereby demonstrating the competence of RoBERTa for semantic representation and feature extraction. (2) In the nested RE task with the Policy data set, the number of sentences processed by our method per second was less than those by the two SOTA baseline models, but the F1 value was much higher with our method than the two baselines, mainly because our method predicted more candidate relations in each layer, which ensured the higher-level relation identification performance, whereas the other two baseline algorithms predicted fewer candidate relations in the lower layers. The calculation speed was faster for the two baseline models but they omitted several potential lower level relations, thereby resulting in errors in higher level RE and the performance of the nested RE task degenerated. (3) In the flat RE task with the KBP37 and SemEval-2010 data sets, the input sentences had no multilayer structure, so the computational efficiency of our method was almost the same as that of the BERT-based BERT+DAG-LSTM method, which again verified the efficiency of our method.

6.9. Case study

Two realistic nested RE examples are presented in Figs. 7 and 8 to compare the performance of our SCL-nestedRE method with the baseline INN method.

According to Fig. 7, our method correctly extracted the nested relations, whereas the baseline INN method missed the relation triple <"*company*", @, "*self-owned capital*"> in the first layer and the nested relations in higher layers that depended on this relation triple as a constituent element. As shown in Fig. 8, the baseline method mistakenly identified time entity "*past three years*" and entity "*sales revenue*" as a relation triple and consequently, redundant entities were generated in higher layers in the nested RE process.

It is possible that the baseline method was not sensitive to the order of entities and it did not fully exploit the structure and semantic information in the input sentence. In input sentence 1, the entity "*self-owned capital*" is in front of the entity "*company*," and in the ground-truth relation triple, the entity "*company*" is placed before the entity "*self-owned capital*." Due to the directional constraints on the relation triple, the baseline method may not have correctly identified the relation triple < "*company*", @, "*self-owned capital*">. Similarly, for input sentence 2 in Fig. 8, the baseline method incorrectly paired two entities "*past three years*" and "*sales revenue*" in a relation triple, whereas our method avoided the generation of redundant relations. Therefore, our method performed well in the nested RE task.

7. Conclusions and future work

In this study, we clarified the definition of the nested RE task and illustrated its potentially wide applicability. We then proposed an INN model based on Transformer for extracting nested relations in a layer by layer manner by taking the types and hidden states of the constituent elements of relation triples as the input feature sequences. Furthermore, to ensure that the proposed model can still work well in low-data settings, we proposed an optimization strategy based on the SCL method. By comparing the structure and semantic similarities between input sentence pairs, this strategy allows our iterative Transformer-based model to learn feature representations that can more effectively express the characteristics of nested relations, and it can also serve as a

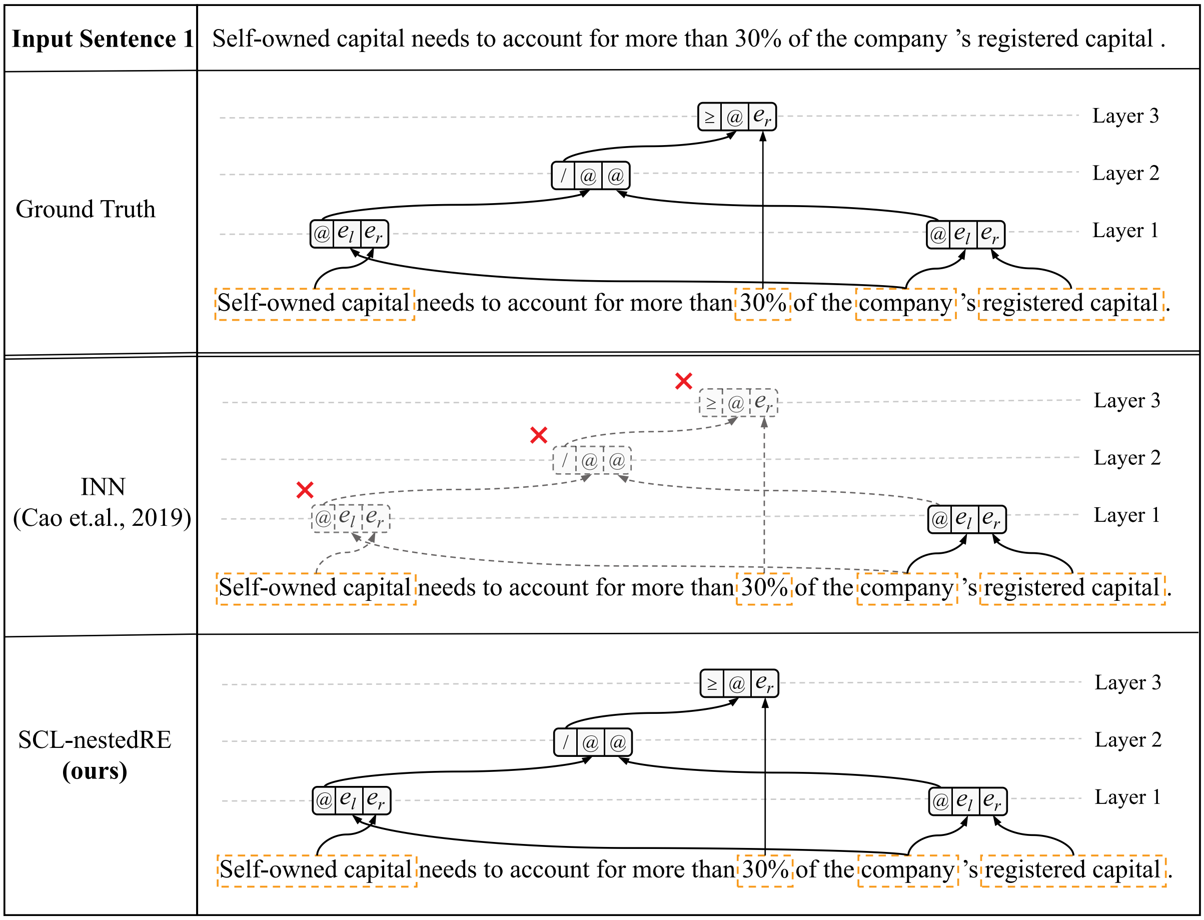


Fig. 7. Example of nested relation extraction from the Policy data set. The baseline method was affected by the relation omission problem. The dashed boxes with red cross marks indicate the missing relation triples.

data augmentation method by using the original data set itself without artificially fabricating positive and negative instances. Extensive experimental results showed that our method outperformed the existing SOTA baselines and ablation experiments also verified the effectiveness of our optimization strategy.

Our nested RE method outperformed the previous SOTA baseline methods but some limitations still need to be addressed.

(1) The performance of the nested RE method depends on the quality of the entities and relations extracted in the lower layers. Thus, any entity and relation recognition errors will be propagated to the nested RE in the higher layers, thereby decreasing the performance of the method.

(2) The nested RE task is affected by a severe low-data problem. Most RE methods focus on relations with the flat structure, whereas there have been few studies of nested RE and supervised data sets for the nested relation identification task are lacking, especially in real application scenarios. In addition, the data annotation process is more complicated for nested relations because multiple relations may be present in one sentence and the nested structures between these relations must be annotated in hierarchical order.

According to the limitations mentioned above, future research could focus on improving the performance of nested relationship extraction tasks by addressing the following issues.

First, to solve the entity and RE error propagation problem from lower layers to higher layers, the entity and nested relations joint extraction method could be enhanced to strengthen the intrinsic connections between entity and relations.

Linguistic knowledge, such as in the dependency parsing task, should also be introduced into the nested RE task to help establish the hierarchical structure of the nested relations under the framework of multi-task learning.

Moreover, to alleviate the low-data dilemma, much research could be conducted to improve the performance and practicability of the nested RE task from the perspectives of algorithm optimization and systems engineering.

From the perspective of algorithm optimization, data augmentation and semi-supervised learning methods could be introduced into the nested RE task to solve the problem caused by the limited scale of labeled data. In addition, methods for alleviating the interference caused by noise in artificially generated data require further research. From the perspective of systems engineering, there is an urgent need to develop an integrated platform that can annotate data and train models for the nested RE task. Most of the existing entity and relation annotation platforms only support labeling for flat relations, and platforms are not available that can support nested relation annotation. In addition, in specific application fields, the establishment of an integrated system for data annotation and algorithm training can rapidly improve the efficiency of model deployment and facilitate the management of the whole process comprising data annotation, algorithm training, algorithm optimization, and model application.

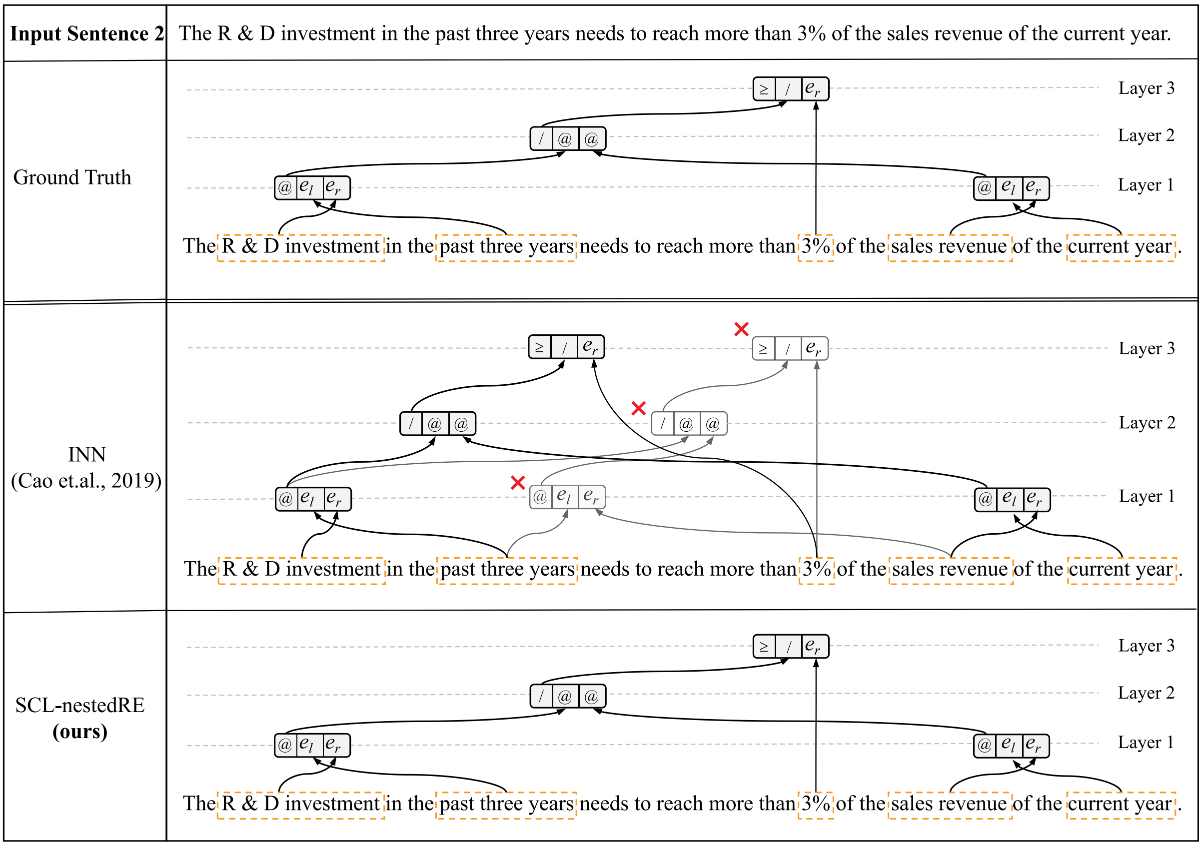


Fig. 8. Example of nested relation extraction from the Policy data set. The baseline method was affected by the relation redundancy problem. The solid boxes with red cross marks indicate the redundant relation triples.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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